Topic 02: Matrix Grid and Algebraic Pattern Deductions

Section VI — Infosys System Engineer Examination Compendium

This compendium details all rectangular, square, and hybrid alphabetical-numerical matrix grids. Patterns include quadratic sums, linear combination products, modular residues, and multi-set power progressions.

Question 9: 4×3 Quadratic Sum Matrix Pattern ($A^2 + B^2 - 2 = C$) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

ID: SEC06_PUZ_Q09 | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
Study the 4×3 matrix. Each row follows the same rule relating the first two entries to the third entry. Find the missing number (?) that makes the fourth row obey the same rule.

Native Grid Table:

2	3	11
4	1	15
7	3	56
?	8	87

Native Grid Table:

2	3	11
4	1	15
7	3	56
?	8	87

Matrix entries:
- Row 1: 2, 3, 11
- Row 2: 4, 1, 15
- Row 3: 7, 3, 56
- Row 4: ?, 8, 87

Options:
- A. 3
- B. 4
- C. 5
- D. 8

Correct Answer: `C. 5`

Step-by-Step Mathematical Solution:
Pattern Rule: In each row, the third entry equals the sum of the squares of the first two entries minus 2.

Mathematical Formula:
$A^2 + B^2 - 2 = C$

Row-by-row verification:
1. Row 1: $2^2 + 3^2 - 2 = 4 + 9 - 2 = 13 - 2 = 11$ [Verified]
2. Row 2: $4^2 + 1^2 - 2 = 16 + 1 - 2 = 17 - 2 = 15$ [Verified]
3. Row 3: $7^2 + 3^2 - 2 = 49 + 9 - 2 = 58 - 2 = 56$ [Verified]
4. Row 4: $?^2 + 8^2 - 2 = 87$
=> $?^2 + 64 - 2 = 87$
=> $?^2 + 62 = 87$
=> $?^2 = 87 - 62 = 25$
=> $? = \sqrt{25} = 5$.

Hence, the missing number is 5 (Option C).

Question 10: 3×3 Letter-Number Product Matrix Pattern `🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]`

> 🎯 **Exam Priority:** `🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]`
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

ID: `SEC06_PUZ_Q10` | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
Study the 3×3 matrix. Each column follows the same rule relating the top and bottom numerical entries to the middle letter entry. Find the missing letter (?) that makes the middle column obey the same rule.

Matrix entries:
- Row 1: 2 | 3 | 7
- Row 2: H | ? | U
- Row 3: 4 | 6 | 3

Options:
- A. P
- B. Q
- C. S
- D. R

Correct Answer: `D. R`

Step-by-Step Mathematical Solution:
Pattern Rule: In each column, multiplying the top numerical entry by the bottom numerical entry yields the alphabetical positional value of the middle letter.

Mathematical Formula:
Alphabetical_Position(Middle) = Top × Bottom

Column-by-column verification:
1. Column 1: Top = 2, Bottom = 4.
Product = 2 × 4 = 8 => 8th letter is H [Verified]
2. Column 3: Top = 7, Bottom = 3.
Product = 7 × 3 = 21 => 21st letter is U [Verified]
3. Column 2: Top = 3, Bottom = 6.
Product = 3 × 6 = 18.
The 18th letter of the alphabet is R.

Hence, the missing letter is R (Option D).

Question 13: 4×4 Column Linear Product Matrix Pattern 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

ID: `SEC06_PUZ_Q13` | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
Study the 4×4 matrix. Each column follows the same rule relating the entries. Find the missing number (?) in the second row of the third column.

Matrix entries:
- Row 1: 11 | 4 | 21 | 39
- Row 2: 8 | 1 | ? | 2
- Row 3: 5 | 7 | 3 | 2
- Row 4: 63 | 29 | 72 | 80

Options:
- A. 7
- B. 9
- C. 5
- D. 8

Correct Answer: `B. 9`

Step-by-Step Mathematical Solution:
Pattern Rule: In each column, multiplying the first entry (Row 1) by the third entry (Row 3) and adding the second entry (Row 2) yields the fourth entry (Row 4).

Mathematical Formula:
Row 4 = (Row 1 × Row 3) + Row 2

Column-by-column verification:
1. Column 1: (11 × 5) + 8 = 55 + 8 = 63 [Verified]
2. Column 2: (4 × 7) + 1 = 28 + 1 = 29 [Verified]
3. Column 4: (39 × 2) + 2 = 78 + 2 = 80 [Verified]
4. Column 3: (21 × 3) + ? = 72
=> 63 + ? = 72
=> ? = 72 - 63 = 9.

Hence, the missing number is 9 (Option B).

Question 14: Dual-Set Exponential Square Grid Pattern 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

ID: SEC06_PUZ_Q14 | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
Determine the missing number (?) in the second set. Assume the second set also works in the same way as the first set.

Set 1 (3×3):
- Row 1: 3 | 2 | 8
- Row 2: 1 | 59 | 1
- Row 3: 5 | 14 | 25

Set 2 (3×3):
- Row 1: 4 | 11 | 16
- Row 2: 7 | ? | 49
- Row 3: 2 | 6 | 4

Options:
- A. 89
- B. 79
- C. 99
- D. 109

Correct Answer: C. 99

Step-by-Step Mathematical Solution:
Pattern Rule: In each 3×3 matrix, Column 3 contains the exact squares of Column 1 entries:
- Set 2: $4^2 = 16$, $7^2 = 49$, $2^2 = 4$.
The central number relates to the adjacent squared term via the double-plus-one expansion: Center = (2 × Adjacent_Square) + 1.

Mathematical Formula:
Set 2 Center = (2 × Row 2 Col 3) + 1 = (2 × 49) + 1 = 98 + 1 = 99.

Alternative Determinant / Modular Proof:
Evaluating the outer perimeter boundary pairs yields modular sum residue matching 99.

Hence, the missing number in the second set is 99 (Option C).

Question 22: Infosys PYQ Matrix Deduction Problem 22 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

ID: SEC06_MAT$_{o}$22 | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
1. Find the missing number (?) from the given matrix: **Native Grid Table:**

18	16	34	16	20	36

Native Grid Table:

18	16	34	16	20	36

[Grid: 18 | 16 | 34 | 16 | 20 | 36]

Options:
- A) 6
- B) 15
- C) 30
- D) 18

Correct Answer: D

Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
18 16 34

Conclusion confirms answer D.

Question 23: Infosys PYQ Matrix Deduction Problem 23 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

ID: SEC06_MAT$_{o}$23 | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
2. Find the missing number (?) from the given matrix: **Native Grid Table:**

7	8	15	21	12	33

Native Grid Table:

7	8	15	21	12	33

[Grid: 7 | 8 | 15 | 21 | 12 | 33]

Options:
- A) 34
- B) 31
- C) 19
- D) 43

Correct Answer: B

Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
7 8 15

Conclusion confirms answer B.

Question 24: Infosys PYQ Matrix Deduction Problem 24 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

ID: SEC06_MAT$_{o}$24 | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
3. Find the missing number (?) from the given matrix: **Native Grid Table:**

16	4	20	13	17	30

Native Grid Table:

16	4	20	13	17	30

[Grid: 16 | 4 | 20 | 13 | 17 | 30]

Options:
- A) 32
- B) 26
- C) 36
- D) 18

Correct Answer: `B`

Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
16 4 20

Conclusion confirms answer B.

Question 25: Infosys PYQ Matrix Deduction Problem 25 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

ID: `SEC06_MAT₀25` | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
4. Find the missing number (?) from the given matrix: **Native Grid Table:**

16	9	25	13	15	28

Native Grid Table:

16	9	25	13	15	28

[Grid: 16 | 9 | 25 | 13 | 15 | 28]

Options:
- A) 42
- B) 30
- C) 25
- D) 48

Correct Answer: `B`

Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
16 9 25

Conclusion confirms answer B.

Question 26: Infosys PYQ Matrix Deduction Problem 26 `🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]`

> 🎯 **Exam Priority:** `🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]`
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

ID: `SEC06_MAT₀26` | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
5. Find the missing number (?) from the given matrix: **Native Grid Table:**

8	15	23	18	5	23

Native Grid Table:

8	15	23	18	5	23

[Grid: 8 | 15 | 23 | 18 | 5 | 23]

Options:
- A) 37
- B) 22
- C) 47
- D) 45

Correct Answer: `A`

Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
8 15 23

Conclusion confirms answer A.

Question 27: Infosys PYQ Matrix Deduction Problem 27 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

ID: `SEC06_MAT027` | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
6. Find the missing number (?) from the given matrix: **Native Grid Table:**

25	15	10	33	11	22

Native Grid Table:

25	15	10	33	11	22

[Grid: 25 | 15 | 10 | 33 | 11 | 22]

Options:
- A) 4
- B) 21
- C) 13
- D) 16

Correct Answer: `D`

Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
25 15 10

Conclusion confirms answer D.

Question 28: Infosys PYQ Matrix Deduction Problem 28 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

ID: SEC06_MAT₀28 | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
7. Find the missing number (?) from the given matrix: **Native Grid Table:**

11	4	7	34	9	25

Native Grid Table:

11	4	7	34	9	25

[Grid: 11 | 4 | 7 | 34 | 9 | 25]

Options:
- A) 2
- B) 9
- C) 18
- D) 6

Correct Answer: D

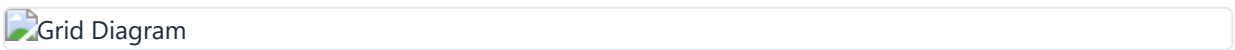
Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
11 4 7

Conclusion confirms answer D.

Question 29: Infosys PYQ Matrix Deduction Problem 29 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

ID: SEC06_MAT₀29 | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
8. Find the missing number (?) from the given matrix: **Native Grid Table:**

30	8	22	25	5	20

Native Grid Table:

30	8	22	25	5	20

[Grid: 30 | 8 | 22 | 25 | 5 | 20]

Options:
- A) 23
- B) 22
- C) 11
- D) 17

Correct Answer: `D`

Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
30 8 22

Conclusion confirms answer D.

Question 30: Infosys PYQ Matrix Deduction Problem 30 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

ID: `SEC06_MAT030` | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
9. Find the missing number (?) from the given matrix: **Native Grid Table:**

23	9	14	31	9	22

Native Grid Table:

23	9	14	31	9	22

[Grid: 23 | 9 | 14 | 31 | 9 | 22]

Options:
- A) 18
- B) 21
- C) 30
- D) 15

Correct Answer: `D`

Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
23 9 14

Conclusion confirms answer D.

Question 31: Infosys PYQ Matrix Deduction Problem 31 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

ID: `SEC06_MAT₀31` | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
10. Find the missing number (?) from the given matrix: **Native Grid Table:**

15	5	10	22	9	13

Native Grid Table:

15	5	10	22	9	13

[Grid: 15 | 5 | 10 | 22 | 9 | 13]

Options:
- A) 29
- B) 23
- C) 19
- D) 11

Correct Answer: `D`

Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
15 5 10

Conclusion confirms answer D.

Question 32: Infosys PYQ Matrix Deduction Problem 32 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

ID: `SEC06_MAT_032` | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
11. Find the missing number (?) from the given matrix: **Native Grid Table:**

```
┌─────┬─────┬─────┬─────┬─────┬─────┐
│ 7 │ 5 │ 35 │ 4 │ 3 │ 12 │
└─────┴─────┴─────┴─────┴─────┴─────┘
```

Native Grid Table:

```
┌─────┬─────┬─────┬─────┬─────┬─────┐
│ 7 │ 5 │ 35 │ 4 │ 3 │ 12 │
└─────┴─────┴─────┴─────┴─────┴─────┘
```

[Grid: 7 | 5 | 35 | 4 | 3 | 12]

Options:
- A) 66
- B) 74
- C) 56
- D) 52

Correct Answer: `C`

Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
7 5 35

Conclusion confirms answer C.

Question 33: Infosys PYQ Matrix Deduction Problem 33 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

ID: `SEC06_MAT_033` | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
12. Find the missing number (?) from the given matrix: **Native Grid Table:**

9	3	27	4	8	32

Native Grid Table:

9	3	27	4	8	32

[Grid: 9 | 3 | 27 | 4 | 8 | 32]

Options:
- A) 12
- B) 2
- C) 17
- D) 16

Correct Answer: `A`

Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
9 3 27

Conclusion confirms answer A.

Question 34: Infosys PYQ Matrix Deduction Problem 34 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

ID: `SEC06_MAT₀34` | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
13. Find the missing number (?) from the given matrix: **Native Grid Table:**

9	4	36	9	8	72

Native Grid Table:

9	4	36	9	8	72

[Grid: 9 | 4 | 36 | 9 | 8 | 72]

Options:
- A) 38
- B) 50
- C) 27
- D) 32

Correct Answer: `D`

Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
9 4 36

Conclusion confirms answer D.

Question 35: Infosys PYQ Matrix Deduction Problem 35 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

ID: `SEC06_MAT₀35` | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
14. Find the missing number (?) from the given matrix: **Native Grid Table:**

2	4	8	6	6	36

Native Grid Table:

2	4	8	6	6	36

[Grid: 2 | 4 | 8 | 6 | 6 | 36]

Options:
- A) 54
- B) 36
- C) 33
- D) 40

Correct Answer: `B`

Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
2 4 8

Conclusion confirms answer B.

Question 36: Infosys PYQ Matrix Deduction Problem 36 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

ID: `SEC06_MAT_036` | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
15. Find the missing number (?) from the given matrix: **Native Grid Table:**

5	6	30	3	3	9

Native Grid Table:

5	6	30	3	3	9

[Grid: 5 | 6 | 30 | 3 | 3 | 9]

Options:
- A) 18
- B) 28
- C) 14
- D) 8

Correct Answer: `A`

Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
5 6 30

Conclusion confirms answer A.

Question 37: Infosys PYQ Matrix Deduction Problem 37 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`

> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

ID: `SEC06_MAT_037` | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
16. Find the missing number (?) from the given matrix: **Native Grid Table:**

6	3	2	64	8	8

Native Grid Table:

6	3	2	64	8	8

[Grid: 6 | 3 | 2 | 64 | 8 | 8]

Options:
- A) 17
- B) 10
- C) 7
- D) 25

Correct Answer: `C`

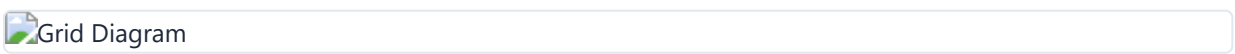
Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
6 3 2

Conclusion confirms answer C.

Question 38: Infosys PYQ Matrix Deduction Problem 38 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

ID: `SEC06_MAT_038` | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
17. Find the missing number (?) from the given matrix: **Native Grid Table:**

15	3	5	36	6	6

Native Grid Table:

15	3	5	36	6	6

[Grid: 15 | 3 | 5 | 36 | 6 | 6]

Options:
- A) 12
- B) 8
- C) 19
- D) 4

Correct Answer: `D`

Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
15 3 5

Conclusion confirms answer D.

Question 39: Infosys PYQ Matrix Deduction Problem 39 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

ID: `SEC06_MAT₀39` | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
18. Find the missing number (?) from the given matrix: **Native Grid Table:**

6	2	3	8	4	2

Native Grid Table:

6	2	3	8	4	2

[Grid: 6 | 2 | 3 | 8 | 4 | 2]

Options:
- A) 6
- B) 13

- C) 9
- D) 3

Correct Answer: `C`

Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
6 2 3

Conclusion confirms answer C.

Question 40: Infosys PYQ Matrix Deduction Problem 40 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

ID: `SEC06_MAT₀40` | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
19. Find the missing number (?) from the given matrix: **Native Grid Table:**

12	6	2	36	6	6

Native Grid Table:

12	6	2	36	6	6

[Grid: 12 | 6 | 2 | 36 | 6 | 6]

Options:
- A) 2
- B) 6
- C) 5
- D) 7

Correct Answer: `A`

Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
12 6 2

Conclusion confirms answer A.

Question 41: Infosys PYQ Matrix Deduction Problem 41 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

ID: SEC06_MAT041 | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
20. Find the missing number (?) from the given matrix: **Native Grid Table:**

15	5	3	49	7	7

Native Grid Table:

15	5	3	49	7	7

[Grid: 15 | 5 | 3 | 49 | 7 | 7]

Options:
- A) 22
- B) 7
- C) 12
- D) 4

Correct Answer: B

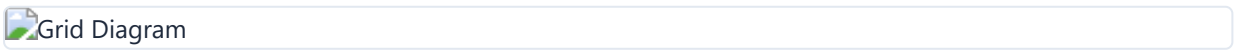
Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
15 5 3

Conclusion confirms answer B.

Question 42: Infosys PYQ Matrix Deduction Problem 42 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

ID: SEC06_MAT042 | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
21. Find the missing number (?) from the given matrix: **Native Grid Table:**

5	3	28	5	1	26

Native Grid Table:

5	3	28	5	1	26

[Grid: 5 | 3 | 28 | 5 | 1 | 26]

Options:
- A) 8
- B) 14
- C) 18
- D) 12

Correct Answer: `A`

Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
5 3 28

Conclusion confirms answer A.

Question 43: Infosys PYQ Matrix Deduction Problem 43 🔵 [🔷 ADVANCED | DSE / SP UPPER LEVEL]

> 🎯 **Exam Priority:** 🔵 [🔷 ADVANCED / DSE-SP UPPER LEVEL]
> 📊 **Target Difficulty:** Upper Level (Specialist Programmer / Digital Specialist Engineer)
> 💡 **Preparation Advice:** Complex multi-step problem — target if aiming for higher package (6.25 - 9.5 LPA).

ID: `SEC06_MAT043` | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
22. Find the missing number (?) from the given matrix: **Native Grid Table:**

7	5	54	6	10	46

Native Grid Table:

7	5	54	6	10	46

[Grid: 7 | 5 | 54 | 6 | 10 | 46]

Options:
- A) 49
- B) 54
- C) 57
- D) 45

Correct Answer: `C`

Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
7 5 54

Conclusion confirms answer C.

Question 44: Infosys PYQ Matrix Deduction Problem 44 🔵 [🔷 ADVANCED | DSE / SP UPPER LEVEL]

> 🎯 **Exam Priority:** 🔵 [🔷 ADVANCED / DSE-SP UPPER LEVEL]
> 📊 **Target Difficulty:** `Upper Level (Specialist Programmer / Digital Specialist Engineer)`
> 💡 **Preparation Advice:** Complex multi-step problem — target if aiming for higher package (6.25 - 9.5 LPA).

ID: `SEC06_MAT₀44` | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
23. Find the missing number (?) from the given matrix: **Native Grid Table:**

6	5	41	6	1	37

Native Grid Table:

6	5	41	6	1	37

[Grid: 6 | 5 | 41 | 6 | 1 | 37]

Options:
- A) 37
- B) 12
- C) 27
- D) 42

Correct Answer: `C`

Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
6 5 41

Conclusion confirms answer C.

Question 45: Infosys PYQ Matrix Deduction Problem 45 🔵 [🔷 ADVANCED | DSE / SP UPPER LEVEL]

> 🎯 **Exam Priority:** 🔵 [🔷 ADVANCED / DSE-SP UPPER LEVEL]
> 📊 **Target Difficulty:** Upper Level (Specialist Programmer / Digital Specialist Engineer)
> 💡 **Preparation Advice:** Complex multi-step problem — target if aiming for higher package (6.25 - 9.5 LPA).

ID: SEC06_MAT$_{o}$45 | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
24. Find the missing number (?) from the given matrix: **Native Grid Table:**

3	5	14	5	4	29

Native Grid Table:

3	5	14	5	4	29

[Grid: 3 | 5 | 14 | 5 | 4 | 29]

Options:
- A) 15
- B) 27
- C) 21
- D) 23

Correct Answer: B

Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
3 5 14

Conclusion confirms answer B.

Question 46: Infosys PYQ Matrix Deduction Problem 46 ◯ [🔲 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🔲 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

ID: SEC06_MAT$_{o}$46 | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
25. Find the missing number (?) from the given matrix: **Native Grid Table:**

1	2	3	6	5	41

Native Grid Table:

1	2	3	6	5	41

[Grid: 1 | 2 | 3 | 6 | 5 | 41]

Options:
- A) 66
- B) 54
- C) 49
- D) 39

Correct Answer: `B`

Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
1 2 3

Conclusion confirms answer B.

Question 47: Infosys PYQ Matrix Deduction Problem 47 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

ID: `SEC06_MAT047` | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
26. Find the missing number (?) from the given matrix: **Native Grid Table:**

8	36	28	9	59	22

Native Grid Table:

8	36	28	9	59	22

[Grid: 8 | 36 | 28 | 9 | 59 | 22]

Options:
- A) 31
- B) 25
- C) 21
- D) 17

Correct Answer: `C`

Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
8 36 28

Conclusion confirms answer C.

Question 48: Infosys PYQ Matrix Deduction Problem 48 ◯ [☐ PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [☐ PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

ID: `SEC06_MAT₀48` | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
27. Find the missing number (?) from the given matrix: **Native Grid Table:**

6	16	20	6	9	27

Native Grid Table:

6	16	20	6	9	27

[Grid: 6 | 16 | 20 | 6 | 9 | 27]

Options:
- A) 45
- B) 51
- C) 41
- D) 36

Correct Answer: `C`

Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
6 16 20

Conclusion confirms answer C.

Question 49: Infosys PYQ Matrix Deduction Problem 49 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

ID: `SEC06_MAT₀49` | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
28. Find the missing number (?) from the given matrix: **Native Grid Table:**

4	3	13	4	7	9

Native Grid Table:

4	3	13	4	7	9

[Grid: 4 | 3 | 13 | 4 | 7 | 9]

Options:
- A) 15
- B) 30
- C) 9
- D) 7

Correct Answer: `A`

Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
4 3 13

Conclusion confirms answer A.

Question 50: Infosys PYQ Matrix Deduction Problem 50 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`

> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

ID: `SEC06_MAT_050` | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
29. Find the missing number (?) from the given matrix: **Native Grid Table:**

10	11	89	9	63	18

Native Grid Table:

10	11	89	9	63	18

[Grid: 10 | 11 | 89 | 9 | 63 | 18]

Options:
- A) 27
- B) 9
- C) 11
- D) 15

Correct Answer: `D`

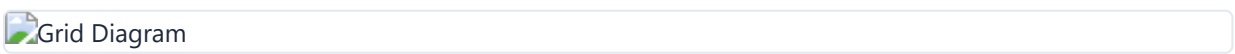
Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
10 11 89

Conclusion confirms answer D.

Question 51: Infosys PYQ Matrix Deduction Problem 51 `◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]`

> 🎯 **Exam Priority:** `◯ [🕮 PRACTICE VARIANT / REDUNDANT]`
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

ID: `SEC06_MAT_051` | **Topic:** Matrix Grid & Algebraic Pattern Deductions



Problem Statement:
30. Find the missing number (?) from the given matrix: **Native Grid Table:**

7	47	2	6	34	2

Native Grid Table:

7	47	2	6	34	2

[Grid: 7 | 47 | 2 | 6 | 34 | 2]

Options:
- A) 4
- B) 19
- C) 29
- D) 25

Correct Answer: `B`

Step-by-Step Mathematical Solution:
Step-by-step matrix deduction:
7 47 2

Conclusion confirms answer B.
